The UML

Grady Booch, as part of his work on OOD in the 1980s, also devised a visual object modeling language. This "Booch method" notation, together with other similar methods, was standardized in the 1990s as the Unified Modeling Language (the UML). Today, the UML is an ISO standard (ISO/IEC 19505). It consists of a set of notation standards covering behavior, interaction, and structure. First, we'll focus on structure.

Classes and Interfaces

The following figure is a UML diagram of how a `Book` class could be built.

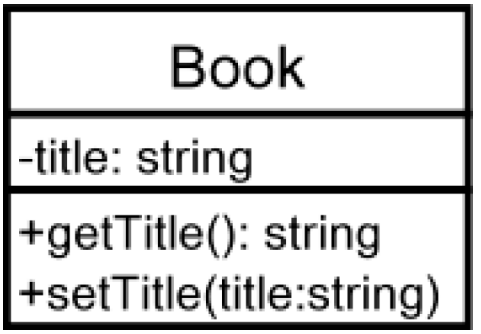


Figure 10-1. *A UML class*

A quick explanation: classes are represented by boxes, which can be sub-divided into up to three sections. The name of the class is always at the top. The section below contains the class "attributes" (a.k.a. properties); data types are optionally shown after the attribute name (names and types are separated by a colon). The final section contains the "operations" (methods). Operations always have parentheses after their names and may show parameters inside the parentheses and return types.

The figure above, therefore, shows that the `Book` class has one private string property called `title` and two public getter/setter methods. `getTitle()` returns a string, and `setTitle()` accepts a string parameter. Members (properties/attributes and methods/operations) have a symbol suffix to denote their visibility: